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COMBINING-ABILITY ANALYSIS AMONG KENYAN AND CIMMYT MAIZE GERMPLASM MID-ALTITUDE ZONE OF KENYA

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Increase in maize yields has been reported where exotic and adapted populations of maize have been crossed; this has been due to the of genetic diversity of the populations involved (Crossa *et al.*, 1987; Mungoma and Pollak, 1988; Darrah *et al.*, 1992; and Vasal *et al.* 1992). Some reports indicate that no immediate positive benefits are obtained from crosses of exotic and adapted maize germplasm (Kramer and Ullstrup 1959; Efrom and Everett, 1969; and Lonnquist, 1974). When heretotic patterns are important but the materials are not adapted, it is possible to adapt and exploit them by including them in crosses with native materials through several back-rosses (Gerrish, 1983). Plant breeders are, therefore, faced with a task of identifying parents that, when crossed, will express maximum heterotis.

In Kenya, herotic patterns in maize have been identified for maize grown in unimodal rainfall areas; however, no similar studies have been undertaken in the mid-altitude, bimodal rainfall zones in western Kenya. The concept of combining ability was developed by Sprague and Tatum (1942). General combing ability (GCA) measures the average performance of a line in a series of hybrid combinations, while specific combining ability (SCA) measures the deviation of certain expected combinations on the basis of average performance of the lines involved. Griffing (1956) proposed the statistical concept of GCA and SCA in homozygous and heterozygous base populations. Diallel analysis allows estimation of GCA and SCA effects and their relative importance in determining the hybrid performance (Matzinger and Kempthorne, 1956; Gilbert, 1958; Gardner and Eberhart, 1966; Baker, 1978).

This study was conducted to determine combining ability effects of selected maize populations and their crosses for grain yield, days to 50% pollen shed and silking, and plant ear height, and to identify heterotic groups of maize populations for future development of hybrid maize in mid-altitude bimodal rainfall zones of western Kenya.

MATERIALS AND METHODS

Six maize populations, three from the Kenya mid-altitude maize breeding programme and three from the International Centre for Maize and Wheat Improvement (CIMMYT) maize breeding programme were selected for diallel study (Table I). They were crossed in a diallel mating system at Kakamega, Kenya, in 1990. For each of the 15 crosses, a minimum of 150 plants from each parental population were used to minimise random sampling variation that is normally associated with small samples.

The parents and their crosses, totalling 21 genotypes (the term genotype is used loosely without implying homozygosity of the populations involved) were evaluated in 8 environments in Kenya during 1992-93 (Table II). The experimental design was a randomised complete block, with four replications in each environment.

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TABLE I - DESCRIPTION OF THE SIX PARENTAL MAIZE POPULATIONS USED IN THE STUDY

Populatuion	Abbr.	Description
Buyore	Byr	The population has a mixture of flint and dent kenels. It is tall and resistant to Northern leaf blight (*Exserohilum turcicum* Pass.) Leonard and Suggs.), and common rust (*Puccinia sorghi* Schwein). Collected from farmers' fields and has been maintained by mild selection at the Regional Research Centre at Kakamega, Kenya
Embu 11	E11	A white endosperm dent breeding population maturing in 150 days at Embu
Embu 12	E12	A white endosperm flint breeding population maturing in 150 days at Embu
Population 21	Pop 21	A white endosperm population selected from Tuxpenno-1 in Zimbabwe by CIMMYT maize programme. It matures in 160 days at Kakamega. In 1988, its performance at Kakamega was 83% of hybrid H625, a leading Kenyan double-cross hybrid that matures in 180 days
Population 29	Pop 29	A population selected in Venezuela from the Tuxpenno Caribe population and improved for grain yield by CIMMYT in Mexico; matures in 160 days at Kakamega. In 1988, its performance was 85% of that of hybrid H625 at Kakamega
Pool 20	P20	A dent grain pool improved for grain yield by CIMMYT in Mexico; it matures in 120 days at Embu

TABLE II - ENVIRONMENTAL MEANS FOR GRAIN YIELD, DAYS TO 50% SILK, DAYS TO 50% POLLEN SHED, PLANT HEIGHT, AND EAR HEIGHT OF 21 MAIZE GENOTYPES USED IN THE STUDY

Location	Altitude (m)	Season	Grain yield (kg/ha)	Days to silk	Days to pollen shed	Plant height (cm)	Ear height
Kakamega	1585	LR	5926	75	72	263	150
		SR	5021	79	78	227	144
Embu	1493	LR	6043	79	78	256	132
		SR	8931	81	80	217	110
Bukura	1463	LR	5044	72	71	201	110
		SR	3544	75	74	195	129
Alupe	1189	LR	4495	67	64	195	104
		SR	4095	69	66	194	105
Mean			5395	76	75	245	147
LSD (0.05)			780	2.3	1.5	2.3	15
CV%			13	3	2	7	9

LR - Long Rains
SR - Short Rain

The experimental unit consisted of two rows each 5 m long spaced at 75 cm. Plots were overplanted and thinned to a plant density of 53,000 ha. Data were recorded for days to silk (days from planting to 50% silk emergence), days to pollen shed (days from planting to 50% pollen shed), plant height (cm) from the soil surface to the base of the tassel), ear height (cm) from soil surface to the top ear node) and grain yield converted to kg/ha.

Statistical Analysis

Analyses of variance (ANOVA) were conducted for grain yield, days to pollen shed, plant height, and ear height using the plot data for each environment separately and then combined across environments. The interaction G x E was used to test main effects and the random error was used to text interactions. Environments were considered as random effects and genotypes as fixed effects. Griffing's (1956) method 2 (parents and crosses), model I (fixed genotypic effects) was used to estimate GCA and SCA effects. Orthorgonal partitioning of the total genotype variance by the least squares method was used to estimate the single degree of freedom comparison of parents versus crosses, which is a test of average heterosis. Griffing's (1956) method 2 model 1 is as follows:

$Y_{ij} = U + g_i + g_j + S_{ij}$ + eijk, where:

Y_{ij} = the effect of the ijth genotype
U = general mean;
g_i = general combining ability effect of effect of the ith parent;
g_i = general combining ability effect of the jth parent;
Sij = specific combining ability effect for the cross between the ith and jth parents; and
eijk = random error

RESULTS AND DISCUSSION

The most favourable environment was Embu with location means for both seasons being higher than those for the rest of the test locations. The study identified Byr and Pop 29 as suitable populations for use in a recurrent selection scheme; E11 and pop 21 formed another promising set with mean high-parent heterosis of 35.1 and 28.0%, respectively.

The combined analysis of variance showed differences among genotype and crosses and parents showed highly significant genotypic differences for grain yield. Differences in days to silk and days to pollen shed were significant at P = 0.05 (Table III). The parents vs. crosses contrast showed non-significant differences for days to silk and days to pollen shed, and ear height. This could be explained by the predominance effect of additive gene action for these traits and hence, low heterosis leading to no differences between the parents and crosses for these traits. Vasal *et al.* (1993a) reported similar results for quality protein maize germplasm. There were highly significant GCA effects for grain yield, days to 50% Silk, days to 50% pollen shed, plant height, and ear height. The SCA effects were only significant (P = 0.05) for grain yield (Table III). These results are in agreement with those reported by Prassad *et al.* (1988) and Vasal *et al.* (1992, 1993a, 1993b) that showed mostly GCA effects for days to 50% pollen shed, days to silk, plant height, and ear height.

A significant genotype x environment interaction was observed for all the traits studied except plant and ear heights. The interactions of GCA and SCA with environment were significant at P = 0.05 for grain yield. The proportion of variance due to SCA x environment interaction was higher than that for GCA x environment interaction for yield. The predominance of SCA x environment interaction is relative to GCA x environment due to the fact that GCA effects are determined by the additive gene action while the SCA effects are as a result of non-additive gene action and that yield is largely governed by non-additive genetic effects (Comstock *et al.*, 1949).

There were significant differences among the six parent populations and crosses for grain yield (Table IV). The highest-yielding cross averaged over environments was Byr x Pop 29 (7378 kg/ha) and the lowest yield was recorded for pop 29 x P20 (6062 kg/ha), implying that Byr and P20 were sources of good and poor yield genes respectively.

TABLE III - COMBINED ANALYSIS OF VARIANCE OF DIALLEL CROSSES AMONG SIX MAIZE POPULATIONS FOR GRAIN YIELD, DAYS TO 50% SILK, DAYS TO 50% POLLEN SHED, PLANT HEIGHT IN EIGHT ENVIRONMENTS IN KENYA DURING 1992-93

Location	df	Grain yield kg/ha	Days to silk	Days to pollen shed	Plant height	Ear height
Environment (E)	7	99500*	3562*	3574*	11560*	11400*
Replications within E	24	19788*	11*	15	725*	675*
Genotypes (G)	20	13500*	43*	59*	189	128*
Parents vs. Crosses	1	27794257*	7	2	1346*	19
Parents (P)	5	1200*	54*	56*	70	67
Crosses (C)	14	1300*	60*	58*	160	153
GCA	5	705847*	77*	68*	969*	470*
sca	15	658815*	3	4	176	51
G x E	140	500*	68*	197*	150	140
P x E	35	500*	45*	81*	108	95
C x E	98	100	55*	38	130	110
GCA x E	35	60500*	18	10	140	103
SCA x E	105	934112*	16	14	100	95
error	481	160	4	5	60	54

* - Significant at P = 0.05

High-parent heterosis for grain yield among the 21 crosses ranged from 15.0 to 35.1%. This relatively low heterosis is typical of varietal crosses and is due to heterozygosity among parental populations.

General Combining Ability Effects

The parent Byr was the best general combiner for grain yield with a highly significant GCA effect (209 kg/ha) over the eight environments (Table V). The parent P20 had a negative significant GCA effect (-364 kg/ha). The only parent that showed a positive and significant GCA effect for time to silk was Pop 29 (0.63 days). For days to pollen shed, Pop 21 and Pop 29 had significant positive GCA effects (Table V). The Kenyan populations had positive GCA effects for plant height while the exotic germplasm had negative and significant GCA effects for this trait but with a sacrifice in yield. This is a useful finding as the Kenyan breeding programme could utilise the introduced germplasm for the reduction of plant height. Ear height GCA effects followed the same trend as the plant height.

TABLE IV - MEANS OF SIX MAIZE POPULATIONS AND THEIR CROSSES ACROSS EIGHT ENVIRONMENTS IN KENYA FOR GRAIN YIELD, HIGH PARENT HETEROSIS FOR GRAIN, DAYS TO 50% SILK, DAYS TO 50% POLLEN SHED, PLANT HEIGHT, AND EAR HEIGHT IN 1992-93

Genotype	Grain yield (kg/ha)	Heterosis for grain yield (%)	Days to silk	Days to Pollen shed	Plant height	Ear height (cm)
Byr x E11	6457	18.3	73	71	236	138
Byr x E12	6582	18.4	74	71	251	142
Byr x Pop 21	7076	25.2	75	74	230	127
Byr x Pop29	7378	35.1	74	73	226	129
Byr x P20	6655	20.6	72	71	222	124
E11 x E12	6699	18.6	73	70	239	137
E11 x Pop21	7268	28.0	74	73	229	127
E11 x Pop29	6756	27.7	74	72	217	120
E11 x P20	6437	31.4	72	70	216	118
E12 x Pop21	7205	18.1	74	73	238	134
E12 x Pop29	7072	28.7	74	73	228	128
E12 x P20	6331	17.1	73	71	224	125
Pop 21 x Pop 29	6902	26.6	76	75	214	117
Pop 21 x P20	6188	15.0	74	73	211	114
Pop 29 x P20	6062	18.2	74	73	207	113
Byr	5604	—	73	71	238	141
E11	5201	—	73	71	222	126
E12	5660	—	73	72	248	146
Pop 21	5658	—	77	76	128	116
Pop 29	4992	—	75	72	211	121
P20	4769	—	73	71	199	105
P20	4769	—	73	71	199	105
Mean	6331	23	75	73	218	123
LSD (0.05)	852	20	15	1.5	21	21
CV %	15	—	3	2	9	12

Specific Combining Ability Effects

Nine crosses showed significant positive SCA effects for grain yield (Table VI). The cross Byr x Pop 29 had the highest SCA effect for grain yield (911 kg/ha) followed by E11 x Pop 21 (781 kg/ha). For days to pollen shed, plant height, and ear height, the SCA effects were generally negative and significant; however, for days to silk, only four crosses had negative and significant SCA effects Table VI). The paucity in number of crosses showing significant SCA effects could be attributed to the predominance of the GCA effects for these traits (Singh *et al.*, 1974; Dhillon and Singh, 1977 and Beck *et al.*, 1990).

TABLE V - ESTIMATES OF GENERAL COMBINING ABILITY EFFECTS AMONG SIX MAIZE POPULATIONS FOR GRAIN YIELD, DAYS TO 50% SILK, DAYS TO 50% POLLEN SHED, PLANT HEIGHT, AND EAR HEIGHT COMBINED OVER EIGHT ENVIRONMENTS DURING 1992-93 IN KENYA

Parents	Grain yield	Days to silk	Days to pollen shed	Plant height	Ear height
Byr	209*	-0.038*	0.038*	8.44*	7.32*
E11	51	-0.25	0.94*	0.81	1.19
E12	100	-0.44*	-0.51*	12.76*	9.32*
Pop 21	193*	0.32	1.89*	-2.19	-3.82*
Pop 29	127	0.63*	0.69*	-7.44*	-4.25*
P20	-364*	-0.69*	-0.76	-12.50*	-9.81*
SE_{gi}	65	0.17	0.14	1.54	1.15
SE_{gi-gj}	100	0.25	0.22	2.38	1.77
LSD (0.05)	197	0.50	0.43	4.66	3.48

* - Significant at P=0.05

TABLE VI - ESTIMATES FOR SPECIFIC COMBONING ABILITY EFFECTS OF CROSSES AMONG 6 MAIZE POPULATIONS FOR GRAIN YIELD, DAYS TO 50% SILK, DAYS TO 50% POLLEN SHED, PLANT HEIGHT AND EAR HEIGHT COMBINED OVER 8 ENVIRONMENTS IN KENYA DURING 1992-93

Genotype	Grain yield (kg/ha)	Number of days to silk	Number of Days to pollen shed	Plant height (cm)	Ear height
Byr x E11	44	0.63	0.19	1.88	4.63
Byr x E12	20	0.32	-0.25	5.44	-0.75
Byr x Pop 21	519*	-0.13	0.38	-1.13	-2.13
Byr x Pop 29	911*	0.75*	0.07	0.13	-0.19
Byr x P20	556*	-0.94	0.50	1.44	0.31
E11 x E12	290	-0.82*	-0.19	1.07	-0.13
E11 x Pop 21	781*	-0.51*	-0.57	6.01	3.50
E11 x Pop 29	454*	0.25	0.63	-1.25	-2.57
E11 x P20	505*	0.00	0.44	0.82	-0.07
E12 x Pop 21	569*	-0.57	-0.50	2.07	1.88
E12 x Pop 29	623*	-0.19	0.19	-2.07	-3.19
E12 x P20	250	-0.38	-0.38	-1.38	-0.69
Pop 21 x Pop 29	361*	-0.13	0.82*	-1.75	-1.07
Pop 21 x Pop 29	14	-0.31	0.25	1.07	1.44
Pop 29 x P20	78	0.07	0.44	1.32	0.88
SE_{sij}	147	0.37	0.32	3.48	2.60
$SE_{sig-sik}$	169	0.36	0.31	3.36	2.51
$SE_{sij-skl}$	256	0.67	0.58	6.29	4.69

* - Significant at P=0.05

SUMMARY

In a study to determine combining ability and heterotic patterns among medium-maturity maize populations from the Kenya and CIMMYT breeding programme, a six-parent diallel involving three kenyan and three CIMMYT maize populations was made in kenya in 1990. The six parents and their 15 crosses were evaluated in eight environments in Kenya during 1992-93. The overall mean yield was 5395 kg/ha and the highest yield (7378 kg/ha) was from Byr x Pop 29 with high parent heterosis of 35.1%, followed by E11 x Pop 21 (7268 kg/ha) with high parent heterosis of 28%. The kenyan accessions were relatively taller than that from CIMMYT. Both GCA and SCA effects played important roles in the expression of grain yield. For days to silk, days to pollen shed, plant height, and ear height, the GCA effects were more important than the SCA effects.

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